

Student 6 **Ashira Bridget Susan S** 192524316RD

5 / 5 submitted

Q1. Harshad number

ANSWER

```
n = int(input())
sum_digits = sum(int(d) for d in str(n))

if n % sum_digits == 0:
    print(n, "is a Harshad number")
else:
    print(n, "is not a Harshad number")

print("\nHarshad numbers from 1 to 500: ")
for i in range(1, 302):
    if i % sum(int(d) for d in str(i)) == 0:
        print(i, end=" ")
```

Q2. GCD

CODE

```
a = int(input())
b = int(input())

while b != 0:
    temp = b
    b = a % b
    a = temp
print("GCD is", a)
```

INPUT

3
6

OUTPUT

GCD is 3

Q3. sum to a target value

ANSWER

```
def find_pairs(lst, target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num
        if complement in seen:
            pairs.append((complement, num))
        seen.add(num)
    return pairs

data = [int(x) for x in input("Enter numbers: ").split()]
t = int(input("Target: "))
result = find_pairs(data, t)
print("Pairs: ", result)
```

Q4. Bitwise Operators

ANSWER

1. a = 0b1101 (Binary literal)
1101 = 8 + 4 + 0 + 1 = 13
2. b = 0o25 (Octal literal)
25 = (2 * 8) + 5 = 21
3. c = 0x1A (Hexadecimal literal)
1A = (1 * 16) + 10 = 26
4. result = (a & b) \ c
a & b -> 13 & 21 = (01101 & 10101) = 00101 = 5
5 | c -> 5 | 26 = (00101 | 11010) = 11111 = 31
5. Bitwise operations on result (31):
result >> 2 -> 31//4 = 7
result << 1 -> -(31 + 1) = -32

Data Types of Variables

a: int
b: int
c: int
result: int

Q5. binary search

CODE

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1
    while low ≤ high:
        mid = (low + high)//2
        if arr[mid] == target:
            return mid
        if arr[low] ≤ arr[mid]:
            if target ≥ arr[low] and target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        else:
            if target > arr[mid] and target ≤ arr[high]:
                low = mid + 1
            else:
                high = mid - 1
    return -1
data = [4, 5, 6, 7, 0, 1, 2]
t = int(input())
result = rotated_binary_search(data, t)
print("Found at: ", result)
```

INPUT

5

OUTPUT

Found at: 1